

Calculator Skills for Papers 2 & 3

Papers 2 and 3 allow a calculator — but most students leave marks on the table by using it slowly or wrongly. These are the techniques that win marks on the calculator papers. Works for Casio fx-83/85 and most scientific calculators.

Why this matters. Paper 1 was non-calculator. Papers 2 & 3 (3 & 10 June) both allow one — and examiners set harder, more realistic numbers *because* you have it. The students who do best aren't just better at maths; they're faster and more accurate with the calculator, so they finish the paper and avoid silly slips. Spend 20 minutes learning the buttons below and you'll save time and marks on every calculator question. **Key models:** these instructions match the Casio fx-83GT / fx-85GT (the most common UK exam calculators); other scientific models are similar.

Before the exam: Always check your specific calculator is on the permitted list for your exam, that it's in **Degrees** mode for trigonometry, and that any previous work is cleared. Practise these techniques on your own calculator before exam day — button layouts vary slightly between models.

1 Set up your calculator correctly

Put it in Degrees mode (not Radians)

GCSE trigonometry is always in degrees. If your answers to sin/cos/tan questions look wildly wrong, this is usually why. Look for a small 'D' at the top of the screen — if you see 'R' or 'G', change it.

SHIFT → SETUP → Degree

Use Math mode (textbook display)

Lets you type fractions, powers and roots exactly as they look in the question, which cuts mistakes. Most exam calculators default to this.

SHIFT → SETUP → MathIO

Clear previous answers before you start

Old values stored in memory can corrupt a calculation. Reset the memory but not the mode settings.

SHIFT → 9 → 2 (or 3)

2 Fractions, powers and roots — fast and exact

Type fractions properly with the fraction button

Don't convert to decimals unless asked. Keeping fractions exact avoids rounding errors that lose accuracy marks.

the a b/c or $\square/\square$ key

Switch between fraction and decimal answers

Many questions want one specific form. This toggles your answer instantly without retyping.

the S-D key

Powers, roots and standard form

Use the power key for indices, and the $\times 10^x$ key (NOT typing 'x10') for standard form so the calculator treats it as one number.

$\times \square$ and $\times 10^x$

Mixed numbers and improper fractions

Convert between them for the form the question asks for.

SHIFT → S-D

3 Trigonometry without errors

Finding a side: type the equation directly

For $\sin/\cos/\tan$ of a known angle, just type e.g. $8 \times \sin(35)$ — the calculator handles it in one go. Make sure you're in Degrees mode.

$\sin \cos \tan$

Finding an angle: use the inverse functions

When you know the ratio and want the angle, use $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ (the SHIFT versions). e.g. $\sin^{-1}(0.5) = 30^\circ$.

SHIFT $\rightarrow \sin / \cos / \tan$

Sine and cosine rule: work in stages

Type the whole expression where possible, but if it's long, work out the bottom of a fraction first and store it (see memory below) to avoid retyping errors.

4 Memory & ANS — the marks-saving habit

Use ANS to carry the exact value forward

After any calculation, the full unrounded answer is stored in ANS. Use it in the next step instead of typing a rounded number — this is the #1 way students lose accuracy marks.

the ANS key

Store a value in memory for multi-step questions

For compound interest, trigonometry or any multi-stage problem, store an intermediate result and recall it, rather than rounding and retyping.

STO $\rightarrow$ M / RCL $\rightarrow$ M

Never round until the final answer

Round only at the very end, to the accuracy the question asks (e.g. 'to 3 significant figures'). Rounding early is the most common avoidable error on calculator papers.

5 Statistics & table modes (big time-savers)

Mean from a frequency table using STAT mode

Instead of working out Σfx and Σf by hand, enter the values and frequencies into the stats table and read the mean directly. Much faster and fewer errors.

MODE $\rightarrow$ STAT (1-VAR)

Use TABLE mode to plot or check graphs

Generate a table of (x, y) values for a function automatically — useful for drawing curves or checking solutions.

MODE $\rightarrow$ TABLE

Check your answers with the equation solver

On capable models, you can verify a quadratic or simultaneous-equation answer — handy if you have time at the end.

MODE $\rightarrow$ EQN (if available)

Exam-day calculator checklist: (1) Degrees mode on. (2) Spare batteries or a spare calculator if allowed. (3) Clear the memory. (4) For every multi-step question, carry the exact value with ANS or memory — don't round and retype. (5) Always give the final answer to the accuracy the question specifies. These five habits routinely save 5–10 marks across the two calculator papers.